02 · ECRFORMER & PATCHGAN TRAINING PIPELINE

Generator Data Flow, Adversarial Critique, and High-Frequency Sharpening

Project: SAR-Guided Satellite Cloud Removal · ECRformer + Conditional PatchGAN
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Document Purpose & Training Architecture

This document covers the complete **model training pipeline**. It explains how 7-channel multi-modal inputs (S_1 SAR, $S_{2,\text{cloudy}}$ optical, and soft mask M) pass through the **ECRformer Generator** (G_θ), how the **11-channel Conditional PatchGAN Critic** (D_ϕ) evaluates local texture fidelity, how **Opaque Cloud Cutout** prevents modality cheating, and **why these methods were chosen over standard alternatives**.

1 The Two-Player Adversarial Architecture

Training operates as a minimax game between two specialized networks:

1. **Generator G_θ (ECRformer-Lite, 3.62M params):** Ingests 7-channel multi-modal stack and predicts unblended clear reflectance $P_{\text{dir}} \in \mathbb{R}^{B\times 4\times 128\times 128}$.
2. **Discriminator D_ϕ (11-channel PatchGAN, 2.77M params):** Ingests 11-channel conditional stack and evaluates local 16×16 spatial patch authenticity.

2 Forward Data Flow Through ECRformer

Stage	Operation / Layer	Input Shape	Output Shape	Core Role in Cloud Removal
0. Input Split	Slicing across channel axis	$[B, 7, 128, 128]$	$S_2 : [B, 4], S_1 : [B, 2], M : [B, 1]$	Isolates optical, SAR radar, and cloud mask
1. Decoupled Stem	Two separate 7×7 Convs	$S_2 : [B, 4], S_1 : [B, 2]$	$F_{\text{opt}} : [B, 32], F_{\text{sar}} : [B, 32]$	Domain-specific low-level feature extraction
2. 4-Stage Encoder	Depth $[2, 2, 1, 1]$	$[B, 32, 128, 128]$	$[B, 256, 16, 16]$	Multi-scale Cross-Modality Attention (CMA)
3. TSA Bottleneck	Time-Space Attention	$[B, 256, 16, 16]$	$[B, 256, 16, 16]$	Global non-local spatial context modeling
4. Symmetric Decoder	Transposed Convs + CBAM	$[B, 256, 16, 16]$	$[B, 32, 128, 128]$	Upsampling with filtered encoder skip connections
5. Residual Head	3×3 Conv + Mean Prior	$[B, 32, 128, 128]$	$[B, 4, 128, 128]$	Outputs unblended direct prediction P_{dir}

Figure 1: Layer-by-layer tensor progression throughout ECRformer Generator.

2.1 Cross-Modality Attention (CMA)

Inside each encoder stage, optical queries (Q) query radar keys and values (K, V):

$$\text{CMA}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{1}$$

This allows cloudy optical regions to dynamically import physical dielectric surface contours from the penetrating SAR signal.

3 PatchGAN Critique & Opaque Cloud Cutout

3.1 Overcoming the "Blurriness Trap"

Standard L_1 and L_2 losses predict the conditional mean $\mathbb{E}[Y|X]$, which turns highways and building blocks into smooth grey smudges. PatchGAN solves this by critiquing high frequencies across 16×16 local patches.

3.2 Opaque Cloud Cutout (65% Modality Dropout)

In thin haze, optical ground lines faintly leak through $S_{2,\text{cloudy}}$. If left unchecked, the model cheats by ignoring SAR. In 65% of training crops, we erase optical ground edges under the mask using 2-pass 15×15 pooling and bright cloud veiling, **forcing the network to learn SAR feature reconstruction**.

3.3 Composite Training Loss

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{L1} + \lambda_{\text{SDFL}}\mathcal{L}_{\text{SDFL}} + w_{\text{adv}}\mathcal{L}_{\text{adv}} + w_{\text{FM}}\mathcal{L}_{\text{FM}} + \lambda_{\text{corr}}\mathcal{L}_{\text{corr}} \tag{2}$$

with $w_{\text{adv}} = 0.04$ and $w_{\text{FM}} = 0.35$.

4 Why We Used These Methods Over Others

1. Why ECRformer over Standard Convolutional U-Nets (e.g. Pix2Pix, DSen2-CR)?

Alternative: Standard U-Net with local 3×3 convolutions.

- **Failure of Alternative:** Local convolutions cannot see beyond 5×5 pixels. Under vast regional cloud cover (> 50 km), convolutions see only white pixels and cannot borrow context from distant clear ground.
- **Why Our Choice Wins:** Self-attention gives ECRformer an infinite receptive field, enabling it to reference clear ground context $10+$ km away.

2. Why an 11-Channel Multi-Modal Discriminator over Optical-Only Critics?

Alternative: Feeding only the generated optical image P_{dir} to the discriminator.

- **Failure of Alternative:** Optical-only critics only verify whether an image looks like Earth terrain; the generator can hallucinate plausible farms where the ground is actually an industrial port.
- **Why Our Choice Wins:** Conditioning on Sentinel-1 SAR radar (11 channels) forces the generated optical textures to strictly obey the physical microwave backscatter of the real ground below.

3. Why Scale-Decoupled Frequency Loss (SDFL) over Standard L_1 Alone?

Alternative: Mean Absolute Error (L_1) across pixels.

- **Failure of Alternative:** L_1 is dominated by low-frequency illumination, rewarding blurry mean predictions.
- **Why Our Choice Wins:** SDFL uses a Laplacian pyramid to isolate high-frequency octaves, penalizing edge gradient errors independently and boosting Detail Ratio from 0.354 to 0.931 (+142% sharper).

5 15-Second Viva Examiner Defense Quick-Sheet

Q: How many parameters are trained in your system?
Answer: The generator contains 3.62 Million parameters ($\approx$ 14 MB), while the PatchGAN critic contains 2.77 Million parameters, enabling efficient training on a single GPU.

Q: What is the purpose of the Opaque Cloud Cutout augmentation?
Answer: It erases leaked optical ground textures in 65% of training crops to prevent the network from taking optical shortcuts, forcing it to utilize Sentinel-1 SAR radar.